

JAYANT TOMAR

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EDUCATION

Delhi Technological University (DTU), New Delhi, India

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.34

2024 – 2028

Class 12th CBSE Board

Percentage: 95%

Class 10th CBSE Board

Percentage: 97.2%

PROJECTS

Fast Mini Search Engine | *C++, BM25, Inverted Index, Trie, Multithreading*

[GitHub](#) | [Video](#)

- Built a **multithreaded C++ search engine** using an inverted index with **BM25** ranking.
- Implemented **Trie-based autocomplete** and **Levenshtein spell correction** to improve search accuracy.
- Designed a C++ HTTP backend exposing **REST APIs** for search, snippets, autocomplete, **document upload**, indexing, search and indexing **latency** and **benchmarking** operations.
- Developed **Hybrid multithreaded indexing pipeline** with incremental **runtime ingestion** and temporary **disk-backed runtime corpus**.
- Implemented **parallel document indexing** using **thread-local inverted indexes** with **post-merge** aggregation, achieving **2.14×** indexing speedup and sub-millisecond average search latency (**0.83 ms**, **1200 queries/sec** throughput).

High-Performance In-Memory Key-Value Store | *C++, TCP*

[GitHub](#) | [Video](#)

- Engineered **thread-safe** operations using **mutex-based** synchronization, ensuring **data integrity** for over 500 concurrent clients with sub-millisecond latency (**0.305 ms** avg, $p99 < 0.9ms$).
- Architected **hybrid persistence (WAL + snapshot/RDB)** enabling crash recovery and fast restart by combining log replay with periodic full-state snapshots.
- Optimized **LRU cache eviction with heap-optimized TTL expiration ($O(\log n)$)**, replacing full scans, sustaining 55K TTL writes/sec with only **0.018 ms** avg latency..
- Developed **master-slave replication with initial synchronization**, where slaves receive snapshot-based state transfer followed by real-time command replication for consistency.
- Achieved **93K+** GET ops/sec and **60K+** SET ops/sec while integrating graceful shutdown and structured logging, and a zero-dependency **WebSocket-to-TCP proxy bridge** for real-time visual telemetry and cluster state exploration.

ACHIEVEMENTS AND AWARDS

- **Top 12 Finalist:** Hack-LLM (24-Hour Hackathon), IIIT Delhi - [Certificate](#)
- **Competitive Programming:** **Knight** (1885) at LeetCode, **Pupil** at Codeforces and **3-Star** at CodeChef.
- **Algorithmic Consistency:** Solved **600+** **DSA problems** on LeetCode, maintaining a rigorous 200-day streak.
- **Goldman Sachs India Hackathon 2026-CS:** Secured All India Rank **1127** - [Leaderboard](#)
- **JEE Advanced & Mains 2024:** Secured AIR 10,987 in JEE Advanced and **99.27** Percentile in JEE Mains (placed in the top 0.73% of over 1.5 million candidates).

TECHNICAL SKILLS

Languages: C, C++, Python, SQL, HTML, CSS

CS Fundamentals: Data Structure, Algorithms, OOPs, Operating System, DBMS, Computer Networks

Developer Tools: Git, Google Colab, Kaggle, VS Code, Visual Studio, MySQL, JupyterLab

LEADERSHIP & CO-CURRICULARS

UGV-DTU (Unmanned Ground Vehicle Team)

Software Division Member

New Delhi, India

Sept 2024 - May 2025

- Collaborated with team members on software components for robotics systems and Contributed to testing and debugging

CERTIFICATIONS

Stanford University Algorithms Specialisation

Machine Learning Specialization by Andrew Ng

Kaggle Intermediate ML

Kaggle Intro to ML

[Certificate](#)

[Certificate](#)

[Certificate](#)

[Certificate](#)